

CH223 Organic chemistry II

Jun Ohata, Ph.D.

Assistant Professor of Chemistry

About the instructor (Jun Ohata)

MAKE metal complexes (B.S., M.S.)

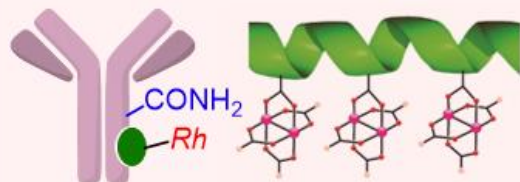
Takemoto, S.; Ohata, J.; Matsuzaka, H. et al. *JACS*. 2014.



Rutheno-
carbene

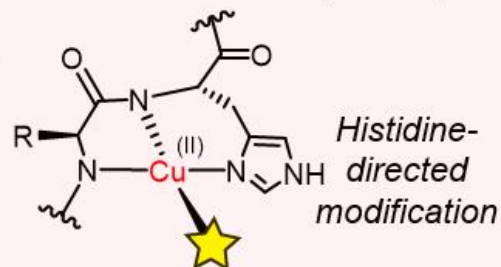


USE metal catalysis for protein modification (Ph.D.)



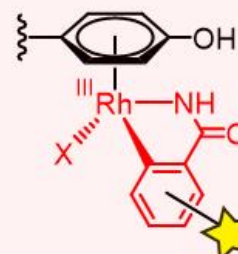
*Rh*₂-proximity-driven reaction

Ohata, J.; Ball, Z. T. *JACS*. 2017.



Histidine-
directed
modification

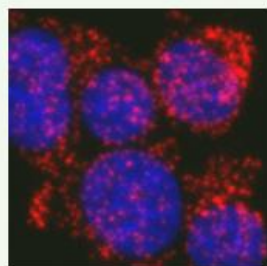
Ohata, J.; Ball, Z. T. et al. *JACS*. 2016.
Ohata, J.; Ball, Z. T. et al. *AC/E*. 2018.



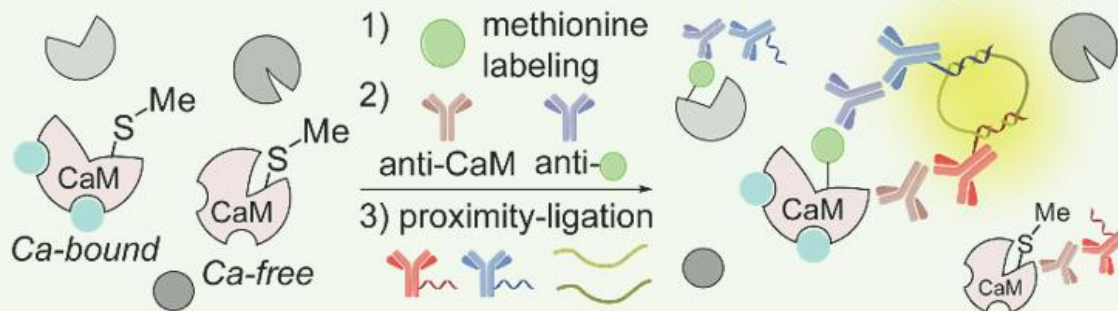
Rh(III)
complexation
on tyrosine

Ohata, J.*; Miller, M. K.*;
Ball, Z. T. et al. *AC/E*. 2018.

SEE metals in living systems (postdoctoral)



Ca imaging



Ohata, J.*; Krishnamoorthy, L.*; Chang, C. J. et al. *ACS Cent. Sci.* 2020.



- Jun received his B.S. in 2011 and his M.S. in 2013 from Osaka Prefecture University (Japan).
- Jun received his Ph.D. in 2018 in Zachary Ball's group at Rice University.
- As a postdoctoral fellow, Jun worked in Christopher Chang's group at the University of California, Berkeley for two years.
- In 2020, Jun moved to North Carolina State University as Assistant Professor (Department of Chemistry).

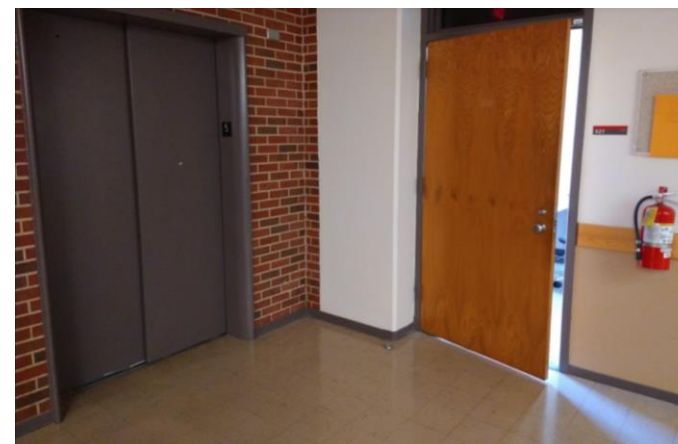
Course instructor

Jun Ohata

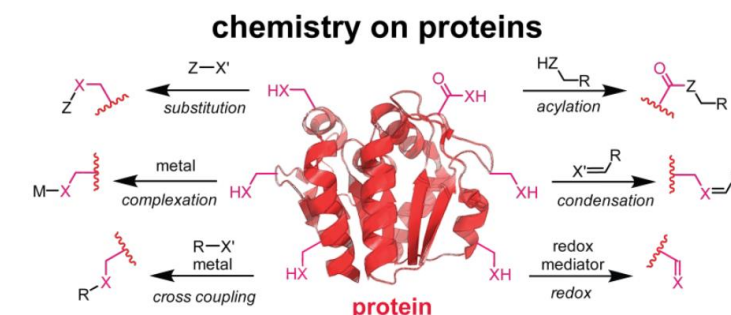
527A Dabney

Email: johata@ncsu.edu

527A Dabney is inside of 527 Dabney. Entrance is near the elevator.



Research laboratory for the Ohata group is also on the 5th floor of Dabney.



<https://ohatalab.wordpress.ncsu.edu/>

Syllabus information: Org II info

North Carolina State University
CH 223, Spring 2023
Organic Chemistry II

Mon-Wed-Fri, 11:45am-12:35pm, in-person lecture at 222 Dabney Hall

Course Description: Second half of two-semester sequence in the fundamentals of modern organic chemistry. Structure and bonding, stereochemistry, reactivity and synthesis of organic compounds. Detailed coverage of aromatic hydrocarbons, conjugated systems and condensation reactions. (Tentative course schedule is available in page 8.)

Learning Outcomes: By the end of the course, students will be able to:

1. Draw structures and name organic compounds including benzene derivatives, carbonyl compounds and nitrogen containing compounds.
2. Describe the attributes that make a molecule aromatic.
3. Describe the bonding, polarity, and the boiling point and solubility trends in the compounds listed in #1.
4. Compare the relative stability and reactivity of various carbonyl compounds.
5. Draw the mechanisms, including transition states and intermediates, and predict the products of electrophilic and nucleophilic aromatic substitution reactions, carbonyl addition/substitution and reduction reactions, carbonyl alpha-C reactions, alcohol oxidation reactions, condensation reactions and saponification.
6. Correlate the results of various organic reactions with predictions from molecular orbital theory.
7. Show the synthetic use of various reagents in functional group transformations.
8. Devise synthetic schemes for preparing various organic compounds.
9. Predict the products for the reactions of various functional groups.
10. Describe the functional groups and reactions of major importance in biochemistry

Attendance for lectures

Attendance: Students are expected to attend all in-person lectures. Missing lecture(s) will put any students at a serious disadvantage with respect to learning course material. It is unnecessary to report the absence for the lectures to the instructor, but if a student wishes to keep the record of the absence, please submit Google Form instead of email the instructor (https://docs.google.com/forms/d/e/1FAIpQLScn9NiD0n8pBa4UK4hf72QaDOtNL-vz4EN2YHcCYLBwqvXZlw/viewform?usp=sf_link).

Course Grading:

Exam I: 200 pts., 20% (02/08/2023, 11:45am-12:35pm)

Exam II: 200 pts., 20% (03/10/2023, 11:45am-12:35pm)

Exam III: 200 pts., 20% (04/12/2023, 11:45am-12:35pm)

Final exam: 250 pts., 25% (5/1/2023, 12:00pm-2:30pm, Dabney 222)

Homework assignments (Achieve): 150 pts., 15% (8 assignments, about 2% each)

- Expected to attend the in-person lectures but attendance is not part of the course grade.
- If you want the instructor to know why you will be absent, please consider using the Google Form, instead of emailing him.

2023. Spring CH223 lecture absence form (not exams)

This Google Form is optional to submit to indicate your absence for lectures. Please use this form rather than email the instructor about your absence. Submission of this form (or emailing the instructor) about your absence for lectures would not affect your course grades.

 johata@ncsu.edu (not shared) [Switch account](#) 

* Required

Your first name *

Your answer

Your last name *

Your answer

Syllabus information: grade and exam

Course Grading:

Exam I: 200 pts., 20% (02/08/2023, 11:45am-12:35pm)
Exam II: 200 pts., 20% (03/10/2023, 11:45am-12:35pm)
Exam III: 200 pts., 20% (04/12/2023, 11:45am-12:35pm)
Final exam: 250 pts., 25% (5/1/2023, 12:00pm-2:30pm, Dabney 222)
Homework assignments (Achieve): 150 pts., 15% (8 assignments, about 2% each)

This course uses standard NC State letter grading:

Points	Grade	Points	Grade
970 +	A+	730 – 769	C
930 – 969	A	700 – 729	C-
900 – 929	A-	670 – 699	D+
870 – 899	B+	630 – 669	D
830 – 869	B	600 – 629	D-
800 – 829	B-	0 – 599	F
770 – 799	C+		

Re-Grade Policy: We are happy to re-grade any question you feel has been improperly graded. However, before turning in your test for a re-grade, please be aware of the following:¶

1) You have seven days from the time the test is returned to request a re-grade. After that time frame has expired, we will no longer entertain re-grade requests.¶

2) To protect the integrity of the class, tests are randomly photocopied and placed on file prior to being returned. If you return an exam for a re-grade in which the answer to the question you want re-graded has been changed in an effort to get additional points, you will be immediately turned over to the office of student conduct for disciplinary action.¶

3) Only exams written in pen can be regraded. Pen copied over in pencil will not be accepted.¶

Missed Exam and Homework Policy:¶

Excuses for *unanticipated* absences must be reported to Dr. Ohata as soon as possible. If you are incapable of taking an exam, a university-approved absence can be used for *ONE* exam (e.g. doctor's note). You must fill out and submit the absence verification form at the bottom of this page: <https://dasa.ncsu.edu/support-and-advocacy/find-help/absence-verification/>. If your absence is verified and approved, you will be given a make-up exam exactly one week after the missed in-class exam during class time. If you miss an exam and do not have a University-approved excuse, you will be given zero for that exam. Excuses for *anticipated absences* must be cleared with Dr. Ohata before the absence. Exams can be taken at an alternate date if 2-week advance notice is provided. Missing more than one in-class exam will result in an incomplete (IN) for this class.¶

University-excused absence information: <https://policies.ncsu.edu/regulation/reg-02-20-03-attendance-regulations/>¶

- Be aware of the grade distribution and the regrading policy!

Course schedule (subject to change)

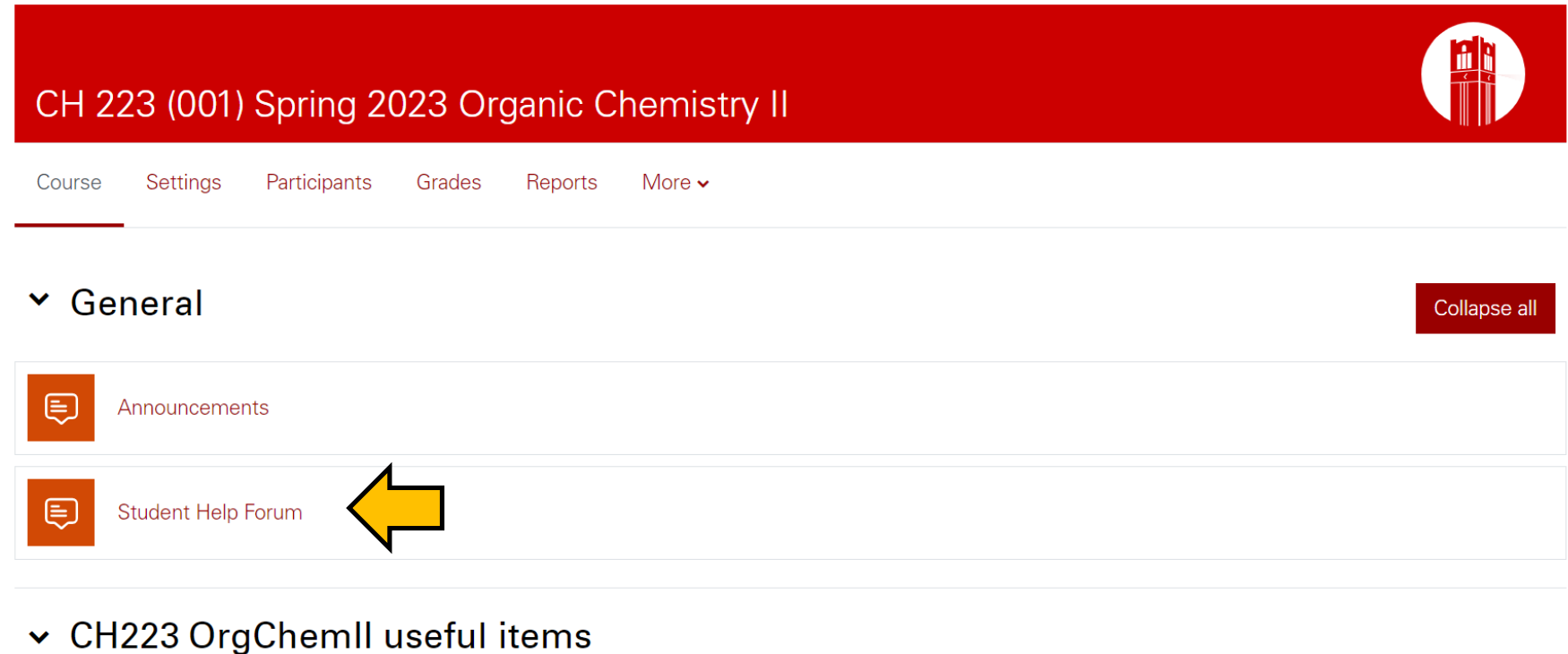
Course Schedule (Subject to Change):

lecture	date			topic	chapter	Loudon
1	Jan	9	Mon	introduction	-	-
2		11	Wed	Dienes, Resonance, and Aromaticity	11	15
3		13	Fri			
		16	Mon	no class, MLK	-	-
4		18	Wed		11	15
5		20	Fri	The Chemistry of Benzene and Its Derivatives	12	16
6		23	Mon			
7		25	Wed			
8		27	Fri			
9		30	Mon			
10	Feb	1	Wed	Allylic and Benzylic Reactivity	12	17
11		3	Fri			
12		6	Mon			
		8	Wed	exam-1	11-12	15-17
13		10	Fri	Aryl Halides, Vinylic Halides, and Phenols. Metal Catalysis	13,15	18
14		13	Mon			
15		15	Wed			
16		17	Fri			
17		20	Mon	Aldehydes and Ketones. Carbonyl-Addition Reactions	18	19
18		22	Wed			
19		24	Fri			
20		27	Mon			
21	Mar	1	Wed			
22		3	Mon	The Chemistry of Carboxylic Acids	19a	20a
23		8	Wed			
24		8	Wed			
		10	Fri	exam-2	13,15,18,19a	18-20a

17		20	Mon	Aldehydes and Ketones. Carbonyl-Addition Reactions	18	19
18		22	Wed			
19		24	Fri			
20		27	Mon			
21	Mar	1	Wed			
22		3	Mon	The Chemistry of Carboxylic Acids	19a	20a
23		8	Wed			
24		8	Wed			
		10	Fri	exam-2	13,15,18,19a	18-20a
		13	Mon	no class, Spring break		
		15	Wed	no class, Spring break		
		17	Fri	no class, Spring break		
25		20	Mon		19b	20b
26		22	Wed	The Chemistry of Carboxylic Acid Derivatives	20	21
27		24	Fri			
28		27	Mon			
29		29	Wed			
30		31	Fri	Enolate Ions, Enols, and α,β -Unsaturated Carbonyl	21	22
31	Apr	3	Mon			
32		5	Wed			
33		7	Fri			
34		10	Mon			
		12	Wed	exam-3	19b-22	20-22
35		14	Fri	The Chemistry of Amines	22	23
36		17	Mon			
37		19	Wed			
38		21	Fri			
39		24	Mon	Course recap	-	-
	May	1	Mon	Final exam (12:00pm-2:30pm, Dabney 222)		

Moodle

- Syllabus and other relevant materials are/will be available.
- Class slides will be uploaded as a pdf file before the class.
- Other announcements will be made through Moodle as well.



CH 223 (001) Spring 2023 Organic Chemistry II

Course Settings Participants Grades Reports More ▾

▾ General Collapse all

Announcements

Student Help Forum

▾ CH223 OrgChemII useful items

Whiteboard images, recorded class video, etc.

Office hour

- The instructor office hours at 527A Dabney: Tue 12:30-1:30 and Thu 12:30-1:30.
- No teaching assistant but please consider using Supplemental Instruction (SI) and Chemistry Tutorial Center (CTC).

SI Leader: Sofia Abello

Course: CH 223 – Section 001

Instructor: Dr. Ohata

Sofia's Session Days and Times

Mondays	6:30 PM – 7:20 PM	Dabney 330
Tuesdays	5:30 PM – 6:20 PM	Dabney 330
Wednesdays	6:30 PM – 7:20 PM	Dabney 330

CH223/224 Tutoring in FOX152

Time	Monday	Tuesday	Wednesday	Thursday	Friday
9:00 - 11:00	Adit Shah	Mohammad Nuruzzaman	Kyle Penn		Toan Ho
11:00 - 1:00	Andrew Hesterhagen	Sanjay Kalliat	Hunter Paggeot		Jared Cossin
1:00 - 3:00	Hayden Mann (2-3)			Nicholas Canty	Michael Kirsch
3:00 - 5:00	Mustapha Abdulmojeed		Chiamaka Uzoewulu	Jonathan Wheeler	

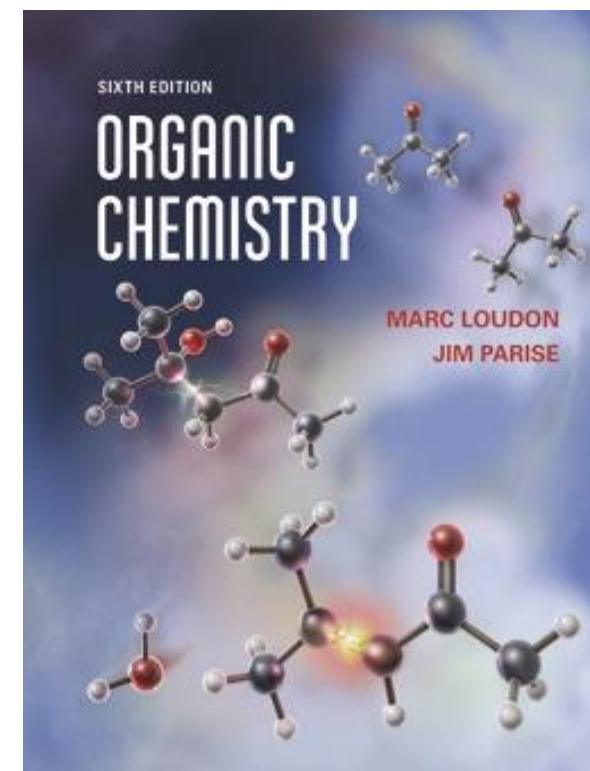
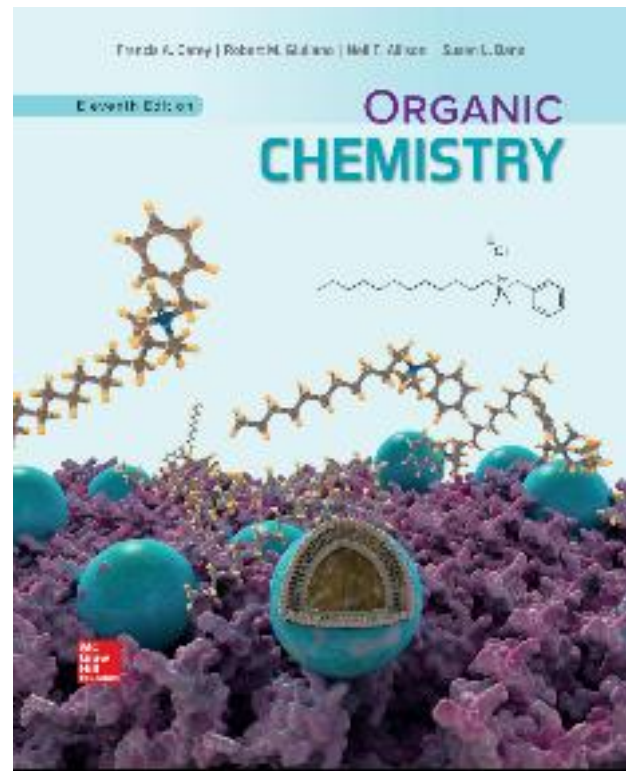
In general, Organic 1 and Organic 2 tutors can answer most questions. If your course tutor is not available, consider visiting a tutor from the other set of courses.

For in-person evening tutoring, please consider the academic success center:

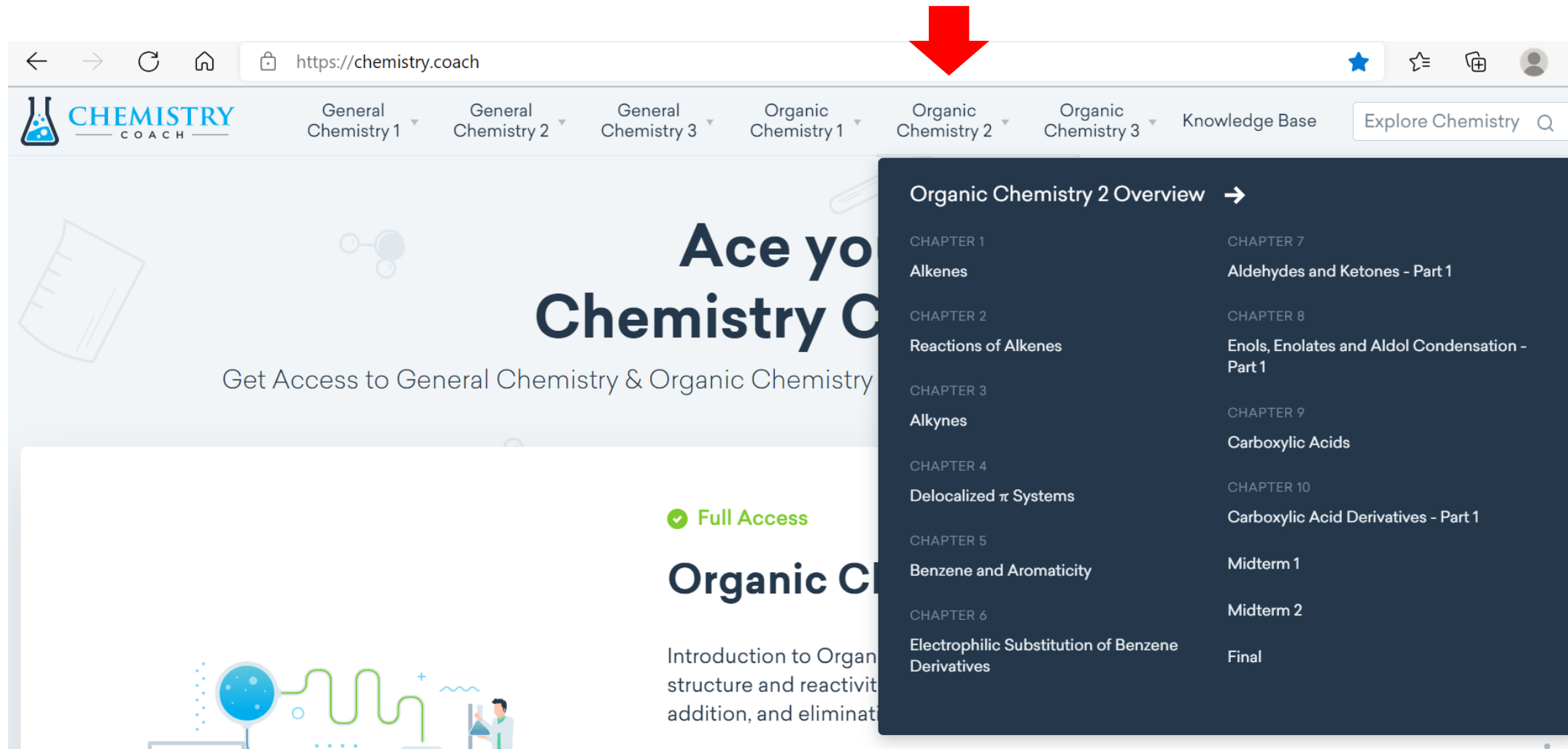
<https://asc.dasa.ncsu.edu/tutoring-and-mentoring/drop-in-tutoring/>

Course materials: textbook

- Carey/Giuliano organic chemistry (11th ed) is a recommended textbook.
- The course slides will be prepared/organized based on Loudon/Parise Organic chemistry 6th Ed.
- The textbook purchase is optional. (free online source, Chemistry Coach will be also used for this course).



Chemistry Coach: Free Online “Textbook”



The screenshot shows the Chemistry Coach website interface. A red arrow points to the 'Organic Chemistry 2' dropdown menu in the navigation bar. The dropdown menu is open, displaying a list of topics for Organic Chemistry 2.

Navigation Bar:

- General Chemistry 1
- General Chemistry 2
- General Chemistry 3
- Organic Chemistry 1
- Organic Chemistry 2** (selected)
- Organic Chemistry 3
- Knowledge Base
- Explore Chemistry

Organic Chemistry 2 Overview

CHAPTER 1	CHAPTER 7
Alkenes	Aldehydes and Ketones - Part 1
CHAPTER 2	CHAPTER 8
Reactions of Alkenes	Enols, Enolates and Aldol Condensation - Part 1
CHAPTER 3	CHAPTER 9
Alkynes	Carboxylic Acids
CHAPTER 4	CHAPTER 10
Delocalized π Systems	Carboxylic Acid Derivatives - Part 1
CHAPTER 5	Midterm 1
Benzene and Aromaticity	Midterm 2
CHAPTER 6	Final
Electrophilic Substitution of Benzene Derivatives	

Chemistry Coach: Free online website

[General Chemistry 1](#) ▼[General Chemistry 2](#) ▼[General Chemistry 3](#) ▼[Organic Chemistry 1](#) ▼[Organic Chemistry 2](#) ▼[C](#)

Pi Bond

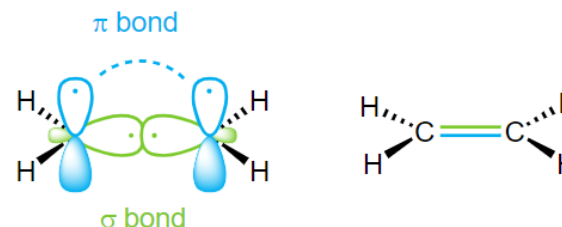
C=C bond of alkenes:

1 σ bond $\Rightarrow$ formed from 2 sp^2 orbitals of carbon atoms

1 π bond $\Rightarrow$ formed from 2 p orbitals of carbon atoms

The π bond is weaker (higher energy) and therefore more reactive than the σ bond

C=C bonds are shorter than C-C bonds



Quizzes and exercises available (optional study)

CHEMISTRY COACH

General Chemistry 1 ▾ General Chemistry 2 ▾ General Chemistry 3 ▾ Organic Chemistry 1 ▾ Organic Chemistry 2 ▾ Organic Chemistry 3 ▾ Knowledge Base Explore Chemistry Q

Home > Organic Chemistry 2 > Benzene and Aromaticity

Chapter 5

Benzene and Aromaticity

Benzene derivatives and aromaticity are studied in this chapter: name and properties of benzene derivatives, Hückel's rule - aromatic, antiaromatic, non-aromatic systems, electrophilic aromatic substitution, halogenation of benzene

[Nomenclature of Benzene Derivatives](#)

[Properties of Benzene](#)

Quizzes ▾

Exercises ▲

Exercise 1

Exercise 2

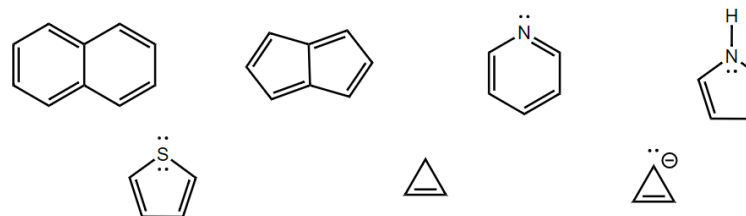
Exercise 3

Exercise 4

Exercise 5

- Chemistry Coach quizzes and exercises are NOT homework but useful resources for your study

Determine if the following molecules are aromatic, antiaromatic or nonaromatic.



See Solutions

Exercise 2

Exercise 3

Exercise 4

Exercise 5

Exercise 6

Exercise 7

Exercise 8

Exercise 9

Exercise 10

Quizzes ▾

How satisfied are you overall to learn chemistry with Chemistry coach?

Achieve for homework

- Required system for the homework.
- ~\$40 for the entire semester.
- Various types of chemistry questions including structure drawing.
- Details available in the last page of the syllabus.

Organic Chemistry

Loudon | Parise SEVENTH EDITION

presented by Macmillan Learning

Show the curved-arrow mechanism for the first step, and the structure of the cyclic intermediate formed, when cyclopentene is treated with KMnO_4 . A Lewis structure for the permanganate ion is provided in the hint. Make sure to show all non-bonding electron pairs and formal charges where necessary. *Omit K^+ .*

Step 1: Draw the curved arrows to form the cyclic intermediate.

Select Draw Rings More Erase

/ // ///

C H Mn O

→

Step 2: Draw the cyclic intermediate.

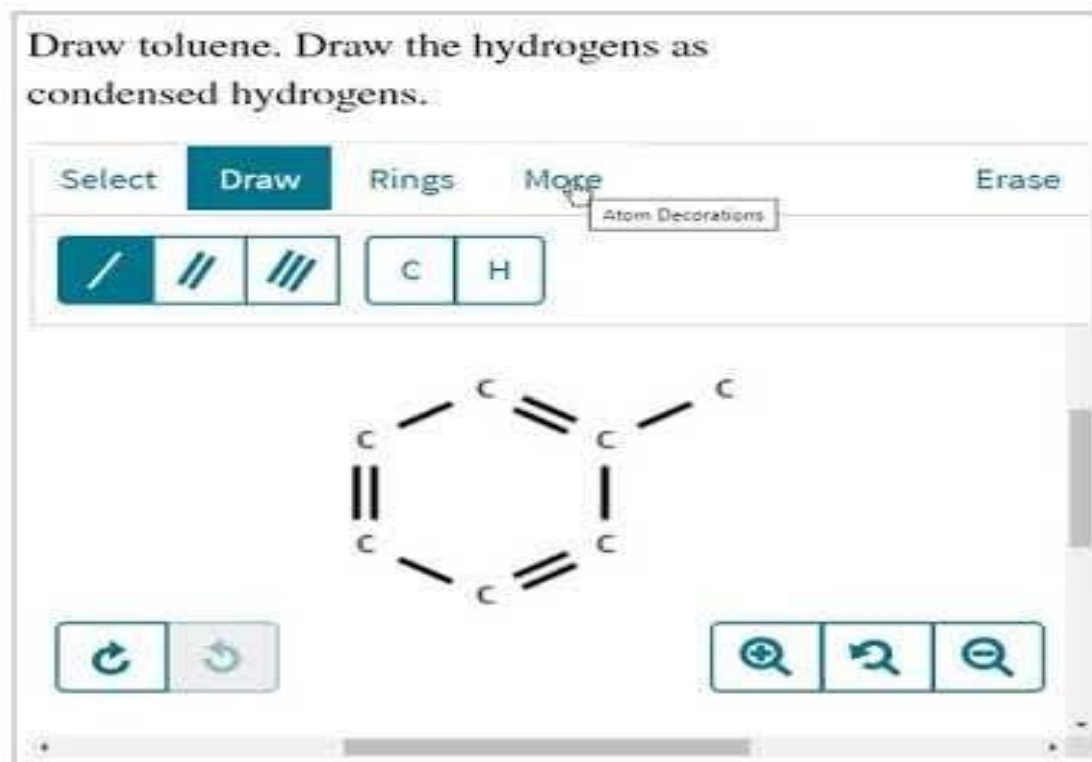
Select Draw Rings More Erase

/ // ///

C H Mn O

→

Achieve for homework

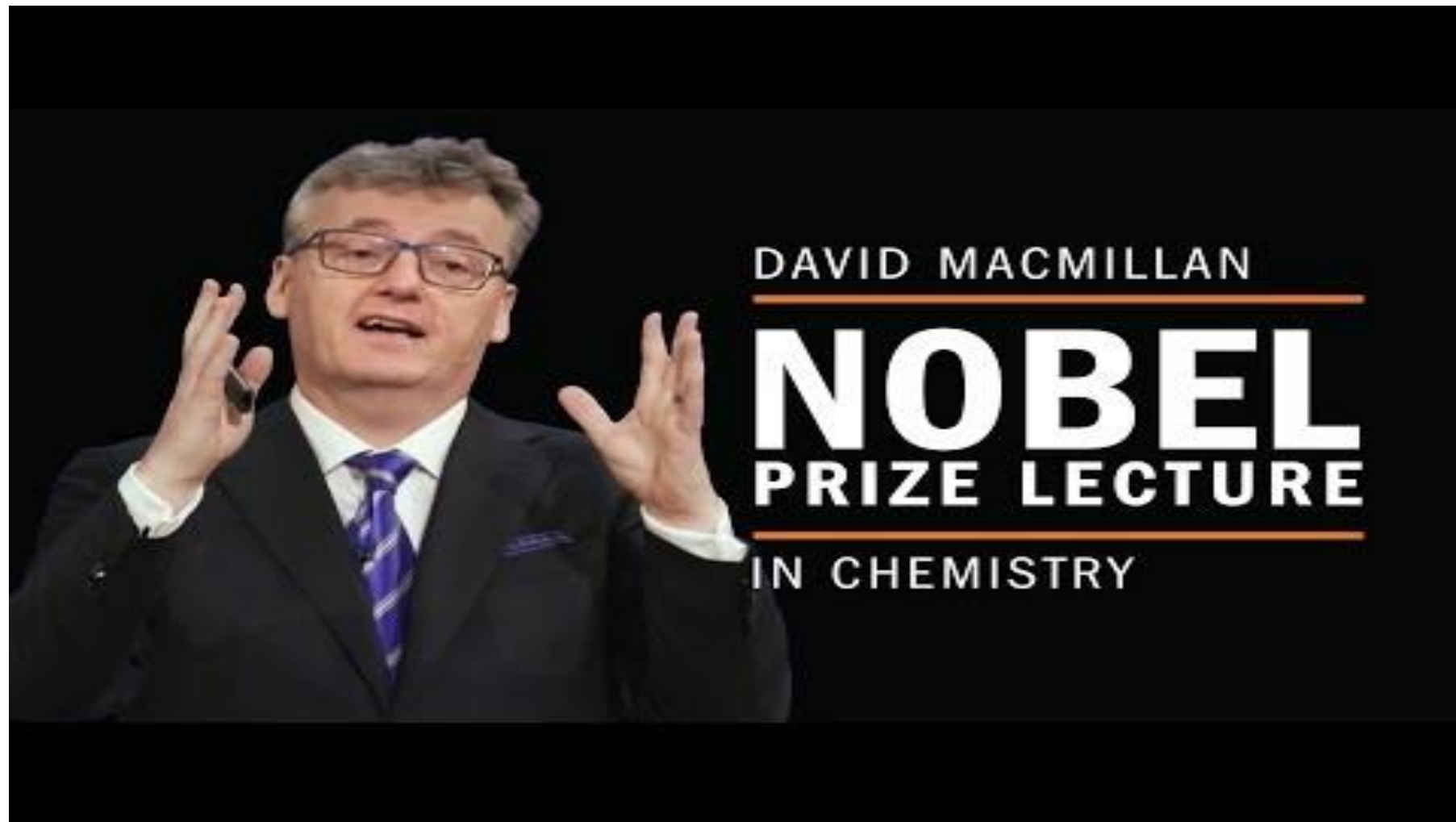


- Helpful tutorial videos available in YouTube.
- Search “Drawing Molecules in Assessments” in YouTube search box, and you will find relevant videos.
- Sometimes, Achieve is quite picky about details. Please be patient with it until you get used to it!

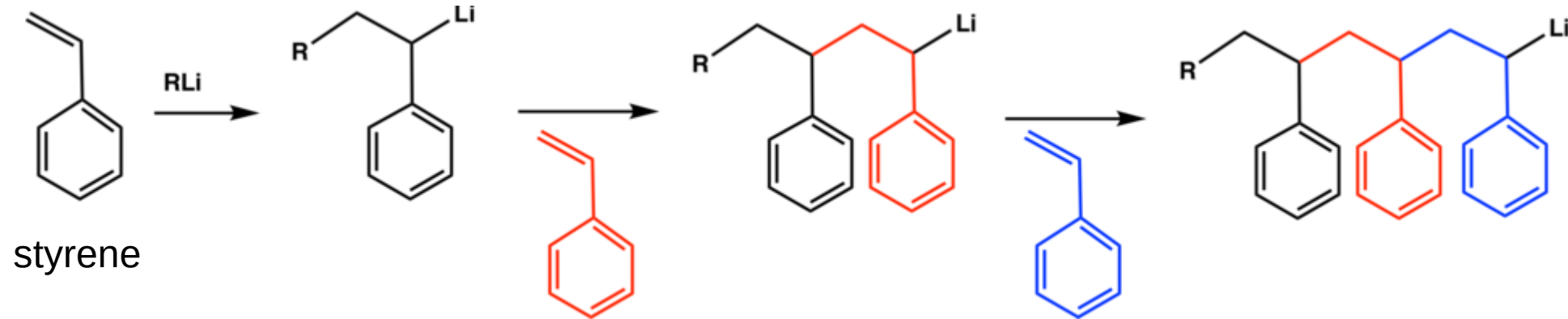
<https://macmillan.force.com/macmillanlearning/s/article/Achieve-Sapling-Learning-Drawing-molecules-in-Assessments-Organic-Chemistry>

Why organic chemistry?

- Nobel Prize 2021. David MacMillan (“for the development of asymmetric organocatalysis”)
- One of the most successful organic chemists on earth at Princeton.



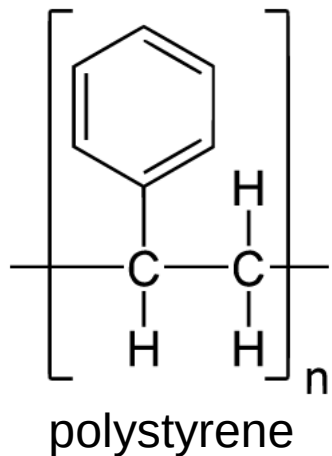
Organic chemistry can make materials (polymer)



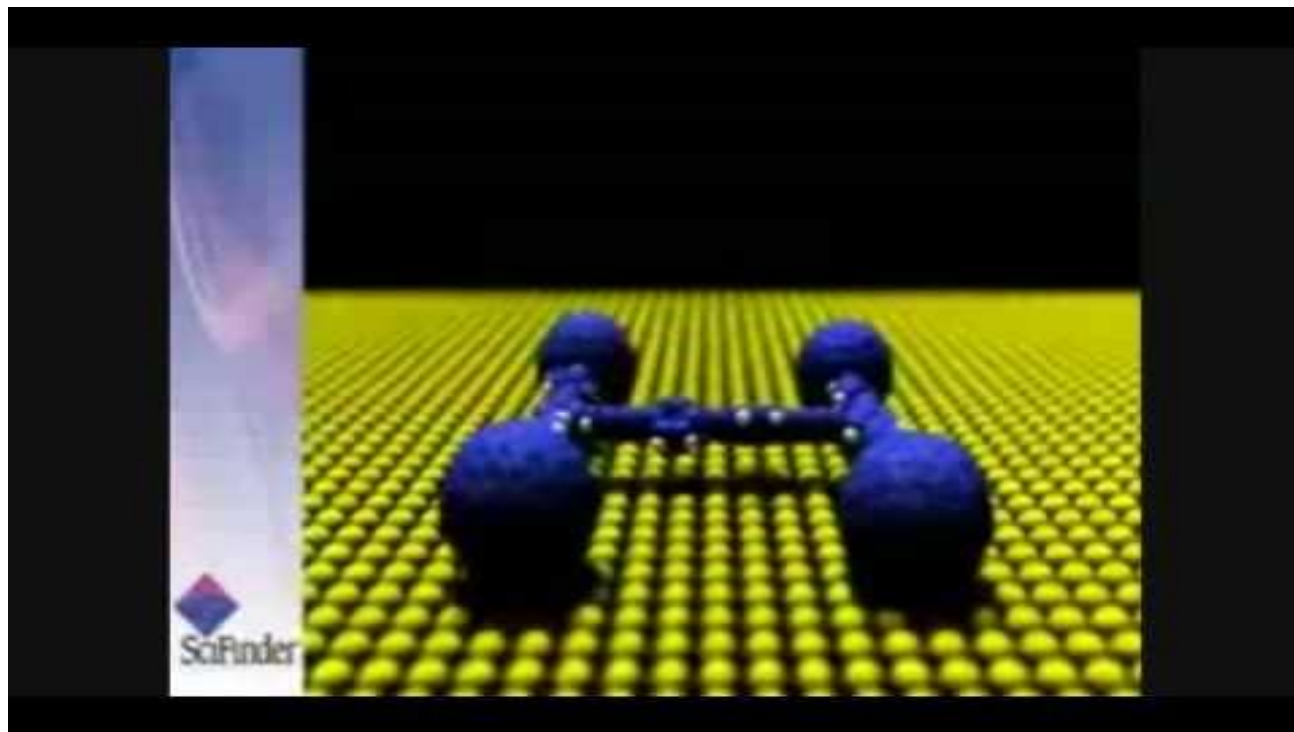
- Polystyrene synthesis
- Combining multiple “styrene” to make polymer.

In the 1940s, researchers, originally at [Dow's](#) Chemical Physics Lab, led by [Ray McIntire](#), found a way to make foamed [polystyrene](#).

<https://en.wikipedia.org/wiki/Styrofoam>

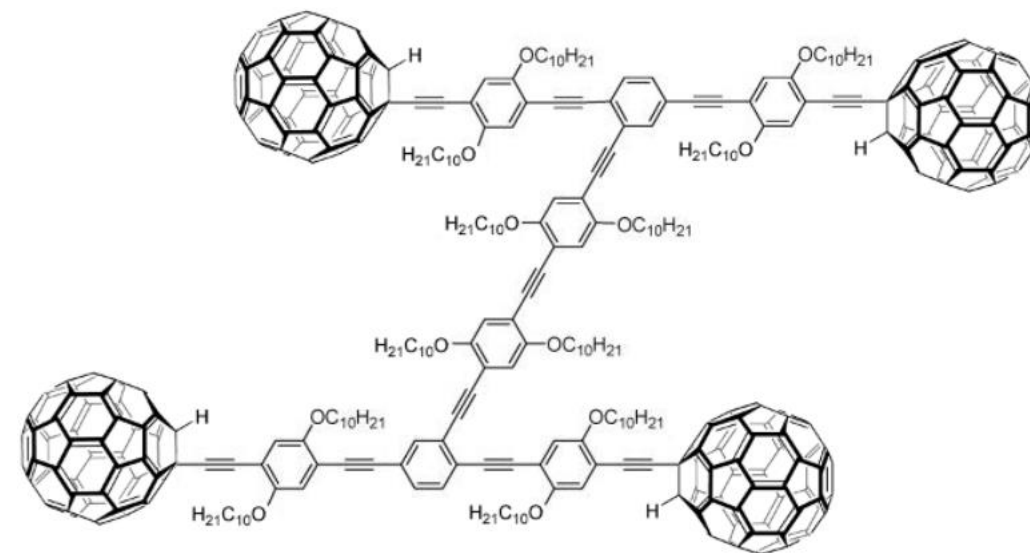


You can even make “nanocar”

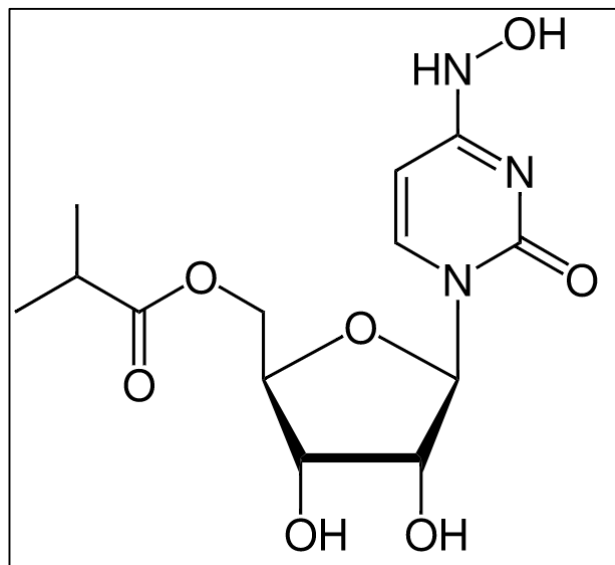


<https://www.youtube.com/watch?v=IASdcW-BtiU>

- Constructing “molecular car” using organic chemistry

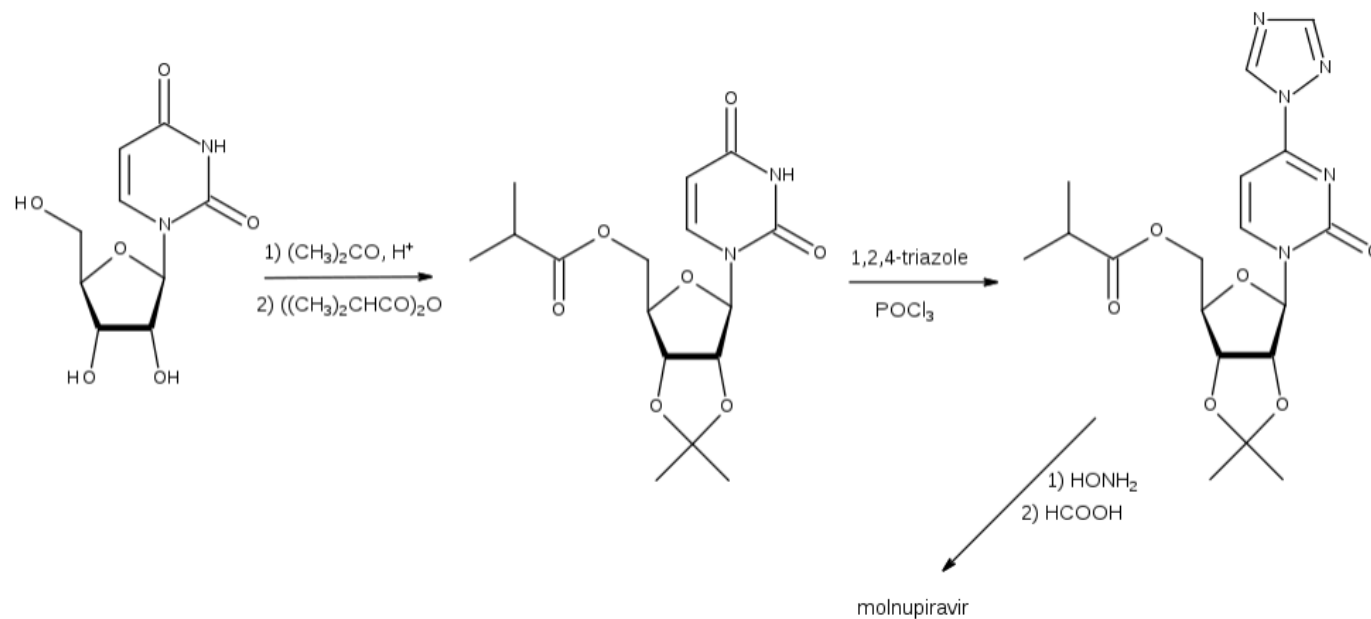


Organic compounds and medicines



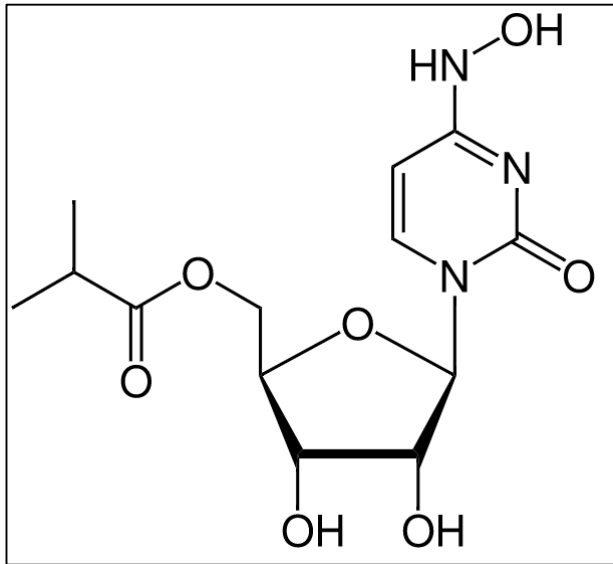
Molnupiravir, sold under the brand name Lagevrio among others, is an antiviral medication that inhibits the replication of certain RNA viruses, and **is used to treat COVID-19 in those infected by SARS-CoV-2.**

Choice of right chemical reactions enabled synthesis of medicines.



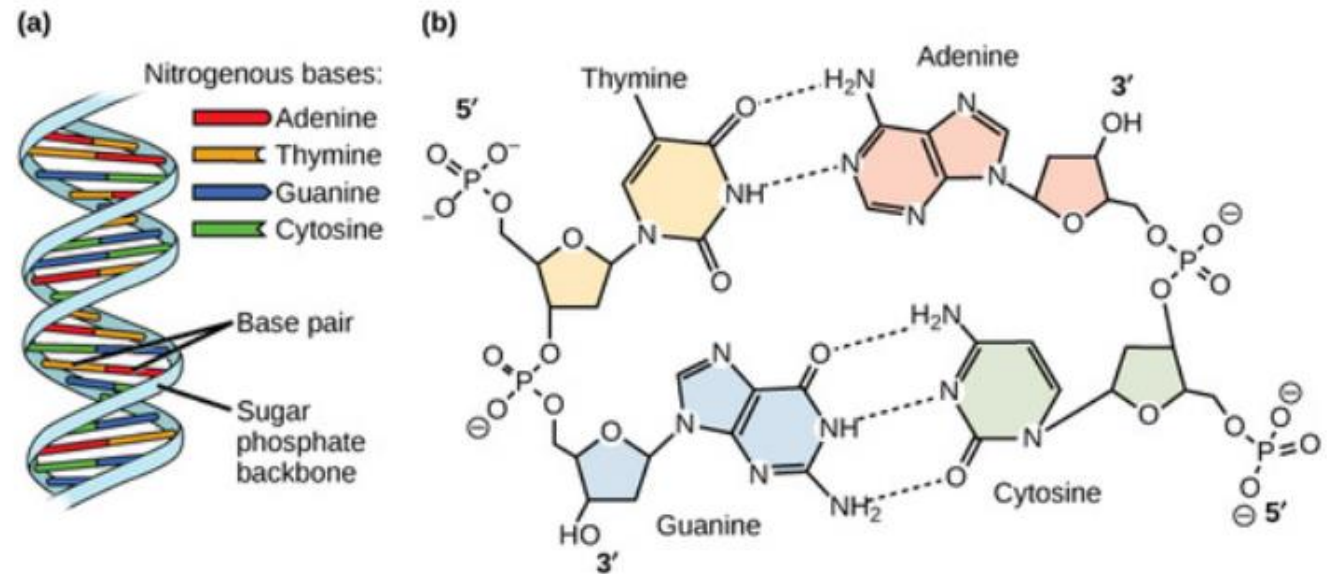
Medicinal chemistry
Total synthesis

Relevance to biochemistry



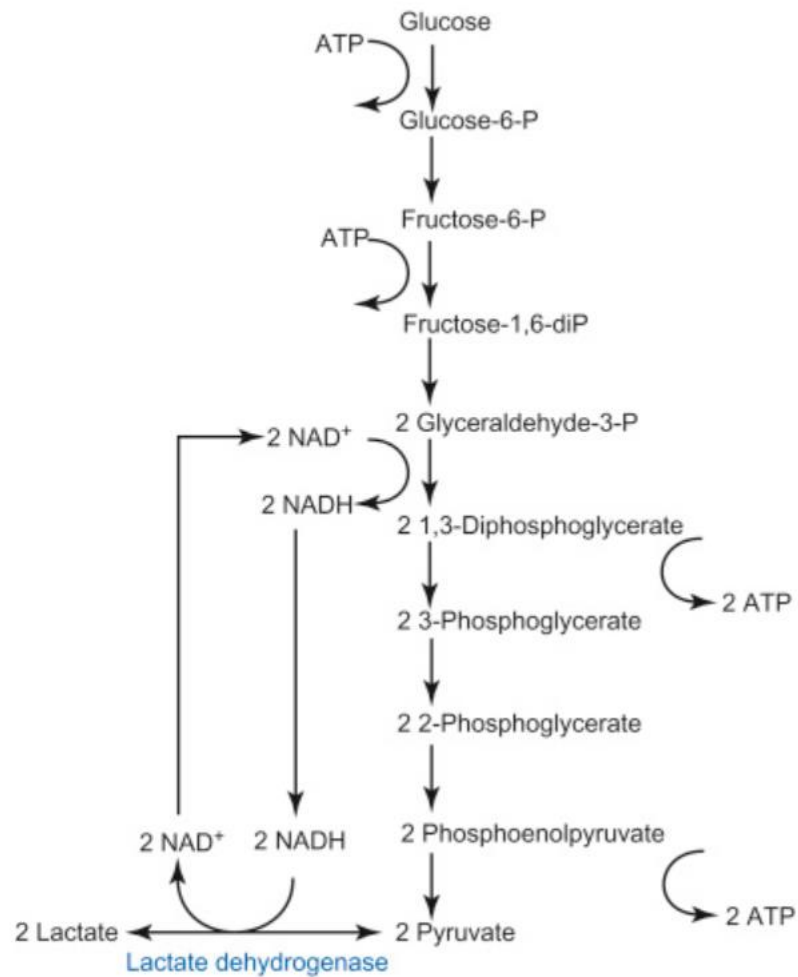
Molnupiravir, sold under the brand name Lagevrio among others, is an antiviral medication that inhibits the replication of certain RNA viruses, and **is used to treat COVID-19 in those infected by SARS-CoV-2.**

Structural similarity between the COVID drug and DNA/RNA structure.
Indeed, the drug targets RNA of viruses.



<https://courses.lumenlearning.com/boundless-biology/chapter/dna-structure-and-sequencing/>

Biochemistry and organic chemistry



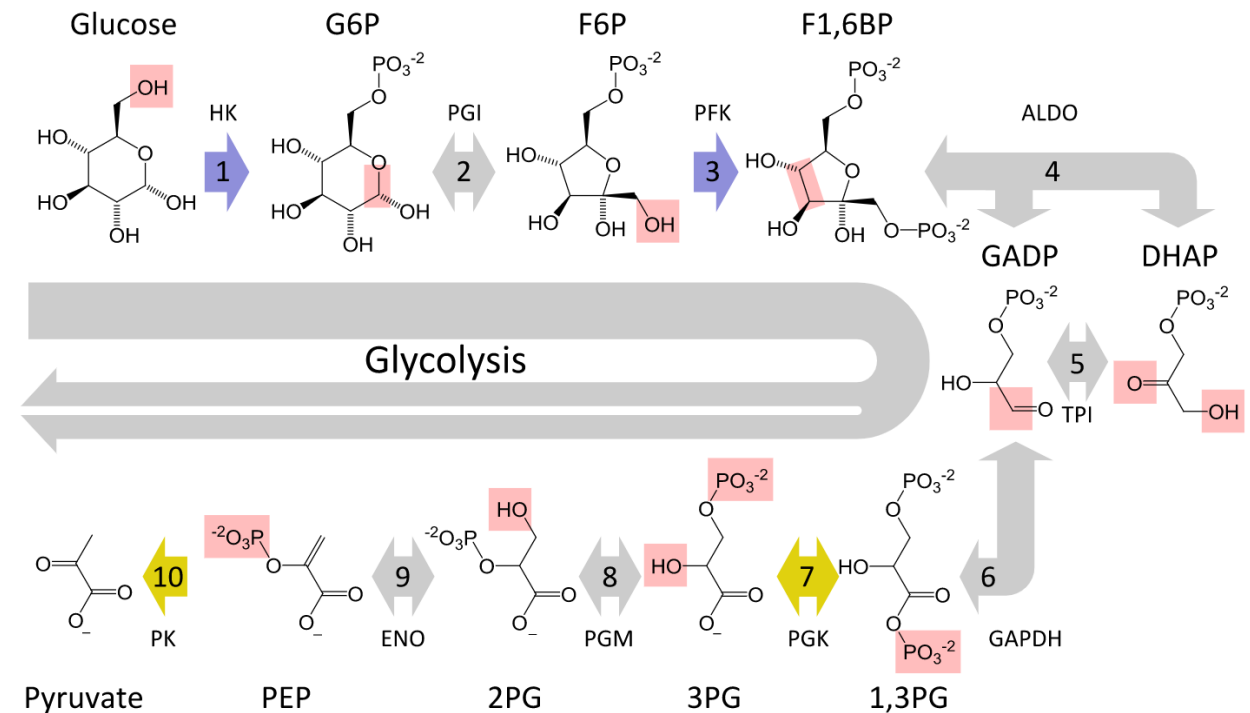
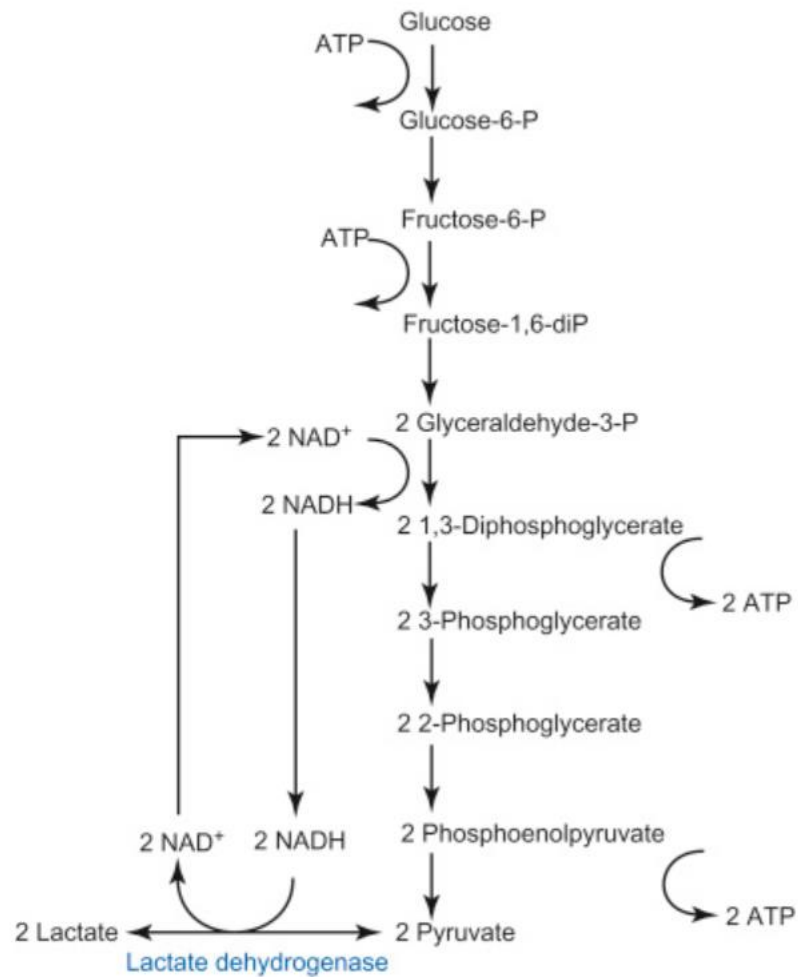
Biochemistry (or biological chemistry)

Biochemistry is **the study of the structure and function of biological molecules** such as proteins, nucleic acids, carbohydrates and lipids. (definition by Nature Portfolio)

You would learn how “chemistry” works in biological systems (example on the left is how cells produce energy from glucose)

Glycolysis: How we generate energy from glucose (sugar)

Biochemistry and organic chemistry



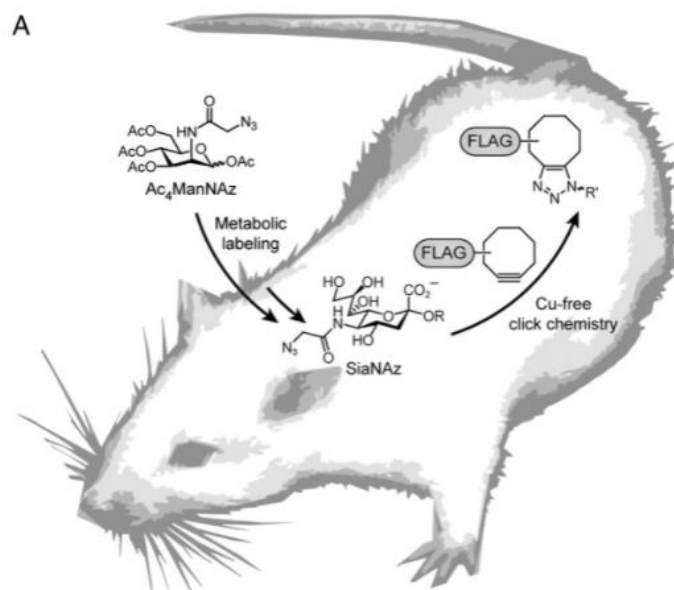
- Depiction by chemical structures (the same materials as the left one!)
- Organic chemistry helps us understand nature

Glycolysis: How we generate energy from glucose (sugar)

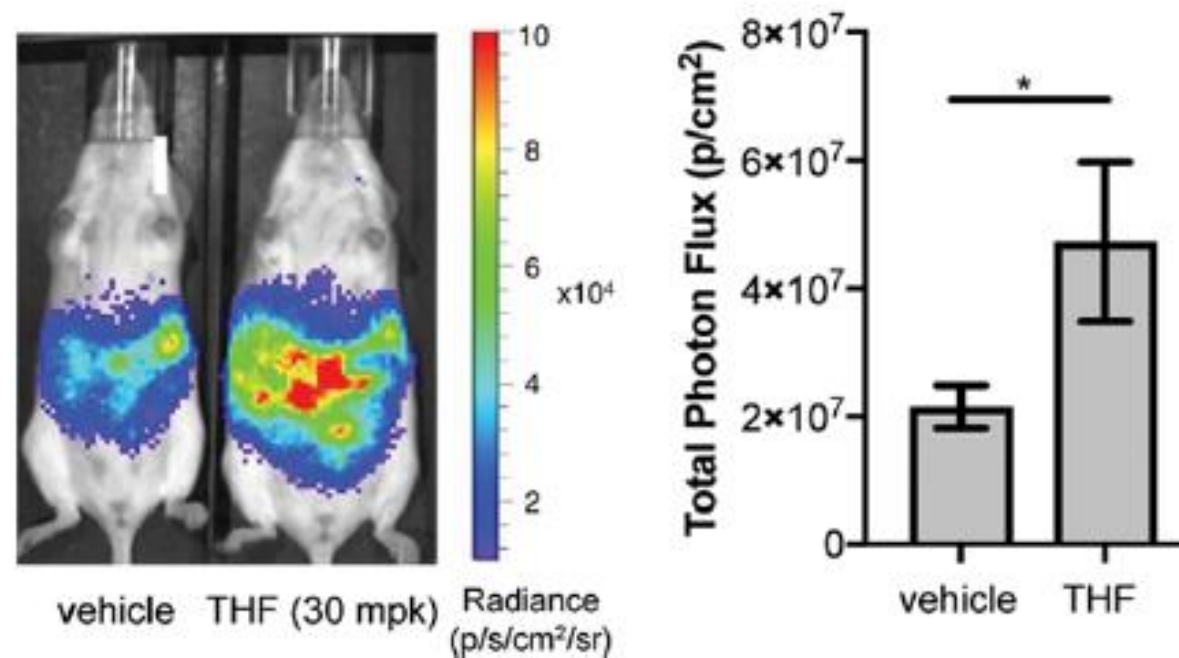
CH795-004 Fall “Chemistry of Life Processes”

Can we do (artificial) organic chemistry in animals?

Organic chemistry can be used to study disease mechanism!



PNAS February 2, 2010 107 (5) 1821–1826



Acc. Chem. Res. 2019, 52, 2841–2848

Org I and Org II

Org II (this course) will be more advanced than Org I (introductory course)



General
Chemistry 1 ▾

General
Chemistry 2 ▾

General
Chemistry 3 ▾

Organic
Chemistry 1 ▾

Organic
Chemistry 2 ▾

Organic
Chemistry 3 ▾

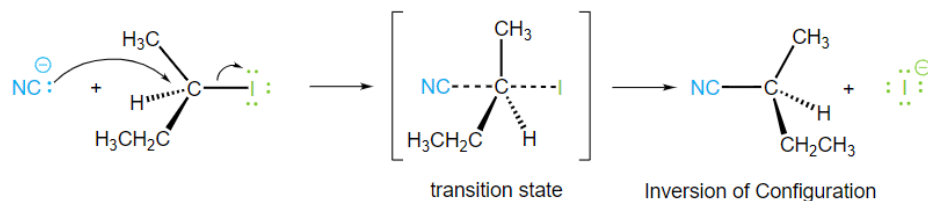
SN2 reactions:

Nucleophilic substitutions which do not proceed via an intermediate $\Rightarrow$ SN2 reactions are bimolecular with simultaneous bond-making and bond-breaking steps. The kinetic rate of a SN2 reaction involves 2 components: the nucleophile and the electrophile reagents

Mechanism:

Because of its unique step, SN2 reaction results in an inversion of configuration at the reaction center. The nucleophile attacks the electrophile center on the opposite side of the good leaving group. The other substituents flip from one side to the other: this is the inversion of configuration

Ex:

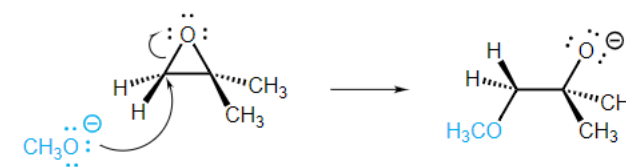


Epoxide opening with strong nucleophiles:

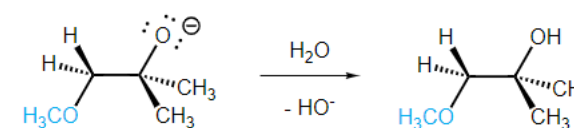
The nucleophile attacks the less hindered carbon atom

Mechanism:

1. SN2 reaction - backside attack



2. Protonation of the alkoxide with water



Org I and Org II

Org I

Brief Contents

	1	Chemical Bonding and Chemical Structure	1
	2	Alkanes	45
	3	Acids and Bases. The Curved-Arrow Notation	87
	4	Introduction to Alkenes. Structure and Reactivity	125
	5	Addition Reactions of Alkenes	181
	6	Principles of Stereochemistry	229
	7	Cyclic Compounds. Stereochemistry of Reactions	272
	8	Noncovalent Intermolecular Interactions	324
	9	The Chemistry of Alkyl Halides	382
	10	The Chemistry of Alcohols and Thiols	452
	11	The Chemistry of Ethers, Epoxides, Glycols, and Sulfides	511
	12	Introduction to Spectroscopy. Infrared Spectroscopy and Mass Spectrometry	569
	13	Nuclear Magnetic Resonance Spectroscopy	611

vi

Org II

	14	The Chemistry of Alkynes	681
	15	Dienes, Resonance, and Aromaticity	712
	16	The Chemistry of Benzene and Its Derivatives	789
	17	Allylic and Benzylic Reactivity	836
	18	The Chemistry of Aryl Halides, Vinylic Halides, and Phenols. Transition-Metal Catalysis	879
	19	The Chemistry of Aldehydes and Ketones. Carbonyl-Addition Reactions	946
	20	The Chemistry of Carboxylic Acids	1004
	21	The Chemistry of Carboxylic Acid Derivatives	1044
	22	The Chemistry of Enolate Ions, Enols, and α,β -Unsaturated Carbonyl Compounds	1103
	23	The Chemistry of Amines	1183
	24	Carbohydrates	1232
	25	The Chemistry of Thioesters, Phosphate Esters, and Phosphate Anhydrides	1283
	26	The Chemistry of the Aromatic Heterocycles and Nucleic Acids	1327
	27	Amino Acids, Peptides, and Proteins	1373
	28	Pericyclic Reactions	1449

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- The materials in Org I and Org II will cover the basics of organic chemistry.
- Mastering those would be like learning how to play chess (but not really good player yet).

Announcements/reminders

- Register/purchase Achieve
- First Achieve homework is due 1/21, 5pm